

EXAM POV NOTES (Q&A FORMAT)

10 Marks Questions

Q1: Dimension Reduction: Detailed Analysis of PCA and LDA, Including Steps and Comparison

Dimension Reduction is the critical process of decreasing the number of features in a dataset while simultaneously retaining essential information. Two prominent methods are **Principal Component Analysis (PCA)**, which is unsupervised and captures variance, and **Linear Discriminant Analysis (LDA)**, which is supervised and maximizes class separation.

- **Dimension Reduction Purpose:** The goal is to simplify models, reduce computational cost, and effectively mitigate the complex challenges posed by the **curse of dimensionality**.
- **PCA Purpose and Use Case:** PCA transforms features into a set of linearly uncorrelated components (principal components) specifically to **capture the maximum variance** in the data. It is suitable for continuous data and operates without needing class labels.
- **LDA Purpose and Use Case:** LDA projects data to a lower-dimensional space in a way that **maximizes class separability**. It is best suited for supervised dimensionality reduction because it requires class labels.
- **PCA Step: Initialisation:** The PCA process starts by **standardising the data** (centering by subtracting the mean) and computing the **covariance matrix**, which measures how features vary together.
- **PCA Step: Component Selection:** PCA identifies components by calculating the **eigenvalues and eigenvectors** of the covariance matrix. Components with the highest eigenvalues are selected because they represent the most variance.
- **LDA Step: Scatter Matrices:** LDA requires computing the **Within-Class Scatter Matrix** (measuring scatter within each class) and the **Between-Class Scatter Matrix** (measuring scatter between different class means).
- **LDA Step: Direction Selection:** LDA solves the **generalized eigenvalue problem** involving the scatter matrices to find the discriminant directions. The top directions are selected to ensure maximum separation between the classes.
- **Comparison: Objective:** PCA reduces dimensionality by **capturing variance**, while LDA's objective is to **maximize class separability**.
- **Comparison: Applicability:** PCA is an **unsupervised** method, while LDA is a **supervised** method that strictly requires class labels.
- **Applications:** PCA is used in applications like **image compression** and feature extraction, whereas LDA is crucial for tasks like **face recognition**, document classification, and medical diagnosis where class separation is key.

Q2: Comprehensive Hyperparameter Tuning: Methods, Controls, and Overfitting Prevention

Hyperparameter Tuning is the optimization process for settings that control the learning algorithm. Hyperparameters (e.g., learning rate) are **set before training**, unlike parameters (e.g., weights) which are learned from the data. This process is essential for achieving optimal model performance.

- **Definition and Distinction:** Hyperparameter tuning optimizes the settings that control the learning process. Hyperparameters (like the number of trees) are set prior to training, while parameters (like neural network weights) are learned during training.
- **Tuning Methods:** Methods used for optimizing hyperparameters include **Grid Search**, which exhaustively evaluates a specific grid of parameters, and **Random Search**, which randomly selects combinations for evaluation. Automated methods like Bayesian Optimization and Genetic Algorithms also exist.

- **Learning Rate:** This hyperparameter controls the **step size** during gradient descent optimization. Small values converge slowly, while large values risk overshooting the optimal minimum.
- **Number of Epochs:** This specifies how many times the **entire training dataset passes through the model**. Increasing epochs can improve training performance but also significantly risks **overfitting**.
- **Batch Size:** This determines the number of training samples utilized in a **single iteration**. Smaller batch sizes provide more updates per epoch but may result in noisier convergence.
- **Dropout Rate:** Used in neural networks, this specifies the **proportion of neurons randomly set to zero** during training, which serves as a vital measure to prevent overfitting.
- **Regularization Parameter (λ / L2 Penalty):** This parameter controls the **magnitude of regularization** applied to model weights (e.g., in Lasso or Ridge regression, or logistic regression). This control is explicitly used to prevent overfitting.
- **Kernel Type:** In Support Vector Machines (SVM), this dictates the kernel function (e.g., linear, RBF, polynomial) used to **transform the input space**.
- **Max Depth:** This hyperparameter limits the depth of trees in decision tree-based models. By constraining the tree size, it acts as a mechanism to **prevent overfitting**.
- **Early Stopping Patience:** This is the specified **number of epochs without improvement** in validation performance that are tolerated before training is automatically halted. It is a critical tool for preventing overfitting.

Q3: Parameter Estimation: Types, Techniques, and the Role of Maximum Likelihood Estimation (MLE)

Parameter Estimation is the process of using observed data to estimate the optimal values for a statistical model's parameters, aiming to best fit the observed data. This process is crucial for creating accurate predictive models and facilitating statistical inference.

- **Definition and Goal:** Parameter estimation determines model parameters that provide the **best fit** for the observed data, effectively making the data most probable. It is used to create accurate predictive models.
- **Frequentist (Point) Estimation:** This type provides a **single value** as the estimate of the parameter, such as using the sample mean to estimate the population mean.
- **Interval Estimation:** This type provides a **range of values** (confidence intervals) within which the true parameter is likely to reside, quantifying uncertainty.
- **Maximum Likelihood Estimation (MLE) Definition:** MLE is a technique that estimates parameters by finding the values that **maximize the likelihood function**.
- **MLE Step 1:** The process involves defining the **likelihood function** based on the observed data.
- **MLE Step 2:** Optimization techniques, such as **gradient ascent**, are used to find the specific parameter values that successfully maximize this likelihood.
- **Bayesian Estimation:** This technique combines existing **prior knowledge** about the parameters with the evidence provided by the observed data to estimate the parameter **distributions**. This is suitable for uncertainty estimation.
- **Statistical Importance of MLE:** Parameters obtained through MLE have clear **statistical meaning** and facilitate crucial statistical processes like hypothesis testing and confidence interval estimation.
- **Use Cases in ML:** Parameter estimation is applied to estimate coefficients in **regression models** (Linear, Logistic), estimate the mean and variance in **Gaussian Mixture Models (GMM)**, and learn weights in **Neural Networks** via backpropagation.

- **Model Selection and Inference:** Parameter estimates support **model selection** by helping compare different models. They also allow for a better understanding of the **relationships between variables** (statistical inference).

5 Marks Questions

Q4: Key Benefits and Techniques of Dimension Reduction

Dimension reduction simplifies models by decreasing the number of features. It is vital for improving model stability and performance in high-dimensional settings.

- **Improved Model Performance:** Dimension reduction enhances performance by eliminating **irrelevant or redundant features**, thus actively reducing noise in the dataset.
- **Mitigating Complexity:** The process reduces computational cost and helps to mitigate the complex challenges associated with the **curse of dimensionality**.
- **Visualization Aid:** High-dimensional data is made much easier to visualize and analyze when reduced, typically down to two or three dimensions.
- **Overfitting Prevention:** By utilizing fewer features, dimension reduction effectively **reduces the overall complexity** of the model, helping to prevent the issue of overfitting.
- **Common Unsupervised Method (PCA):** **Principal Component Analysis (PCA)** is a method that reduces dimensionality by maximizing the variance captured in the data.
- **Common Supervised Method (LDA):** **Linear Discriminant Analysis (LDA)** is a method that maximizes the separation specifically between existing class labels.

Q5: Detailed Steps in Linear Discriminant Analysis (LDA)

LDA is a supervised dimension reduction technique whose primary purpose is to project data to a lower-dimensional space that maximizes the separation between class labels.

- **Within-Class Scatter Matrix:** The first necessary step is to compute the **Within-Class Scatter Matrix**, which quantifies the scatter or dispersion of data points observed *within* each class.
- **Between-Class Scatter Matrix:** Next, the **Between-Class Scatter Matrix** is calculated, which measures the scatter specifically *between* the means of the different classes.
- **Generalized Eigenvalue Problem:** The technique then requires solving the **generalized eigenvalue problem** utilizing these two scatter matrices to determine the discriminant directions.
- **Selection of Directions:** The eigenvectors (discriminant directions) corresponding to the largest eigenvalues are selected, as these directions **maximize the separation** between the classes.
- **Data Projection:** Finally, the original high-dimensional features are transformed and **projected onto the new subspace** defined by the selected discriminant directions.
- **Key Application Suitability:** LDA is highly suitable for tasks where distinct class boundaries are vital, such as **face recognition** and medical diagnosis.

Q6: Key Hyperparameters in Neural Networks and Optimization

Neural networks and optimization processes rely on specific hyperparameters, such as learning rates and weight initialization, which significantly affect convergence speed and overall model performance.

- **Learning Rate:** Controls the **step size** taken during the gradient descent optimization. If too large, it can cause the optimization to overshoot the minimum.
- **Batch Size:** Defines the number of training samples used per iteration, influencing the frequency and **noise level of weight updates**.
- **Dropout Rate:** Specifies the proportion of neurons that are **randomly set to zero** during training, acting as a powerful mechanism to prevent overfitting in the network.

- **Momentum:** In gradient-based optimization, momentum helps to **accelerate gradient descent** by considering the magnitude and direction of previously calculated gradients.
- **Activation Function:** Specifies the function (e.g., ReLU, sigmoid) used within the network to introduce **non-linearity**, enabling the model to learn complex patterns.
- **Weight Initialization:** Determines the method used to initially set the weights in the neural network, which directly affects the model's **convergence speed** and performance.

2 Marks Questions

Q7: Parameter Estimation Definition

Parameter estimation is the process of using observed data to find the optimal values for a model's statistical parameters.

- **Definition:** The goal is determining model parameters that provide the **best fit** for the observed data.
- **Techniques:** Common techniques used include **Maximum Likelihood Estimation (MLE)** and **Bayesian methods**.

Q8: Hyperparameters vs. Parameters

The distinction lies in when the values are set or learned during the modelling process.

- **Hyperparameters:** These are settings (e.g., Batch Size) that are determined and **set before** the training of the algorithm begins.
- **Parameters:** These are values (e.g., weights in a Neural Network) that are **learned** directly from the training data during the optimization process.

Q9: Maximum Likelihood Estimation (MLE)

MLE is a parameter estimation technique focused on maximizing the probability of observing the given dataset.

- **Definition:** MLE estimates parameters by maximizing the **likelihood function**.
- **Importance:** It provides estimates that have **statistical meaning** and facilitates statistical procedures like hypothesis testing.

Q10: Early Stopping Patience

Early Stopping Patience is an optimization mechanism used to control the duration of model training.

- **Definition:** It is the **number of epochs without improvement** in validation performance that are allowed before training is automatically terminated.
- **Function:** Its purpose is to **prevent overfitting** by halting the training process once validation performance stops improving.

Q11: PCA Applicability

PCA is an unsupervised dimension reduction technique widely used for continuous data.

- **Method Type:** PCA is an **unsupervised** method, meaning it does not require class labels to perform dimensionality reduction.
- **Objective:** Its primary objective is to reduce dimensionality by transforming features into components that **capture the maximum variance** in the data.